



DZPHARM • AI LAYER

AI Copilot Core

A detailed deep-dive into the Gemini-powered AI Copilot: trilingual (FR/AR/Darija) with 3 audience modes (professionnel/patient/enfant), structured clinical output (4 sections: rule, calculation, Algerian forms, warnings), grounded in the DzPharm dataset. Three bugs: version label inconsistency (3.8 vs 3.6), suggested prompts don't populate, no conversation history. Plus 11s latency with no streaming.

3LANGUAGES
(FR/AR/DARIJA)**3**

AUDIENCE MODES

3

BUGS

B

GRADE

DEEP-DIVE OF

Feature #36 – AI Copilot Core
From the 68-item review menu

Trilingual, 3-Mode, Structured Clinical Output

The AI Copilot is the platform's **signature innovation** — a Gemini-powered trilingual assistant (French + Arabic + Algerian Darija) with three audience modes (professionnel, patient, enfant). The clinical-grade structured output (4 sections: rule, calculation, Algerian forms, warnings) is genuinely impressive. The system prompt is grounded in the actual DzPharm dataset (9,555 medications, 911 DCI monographs), not generic pharmacology knowledge.

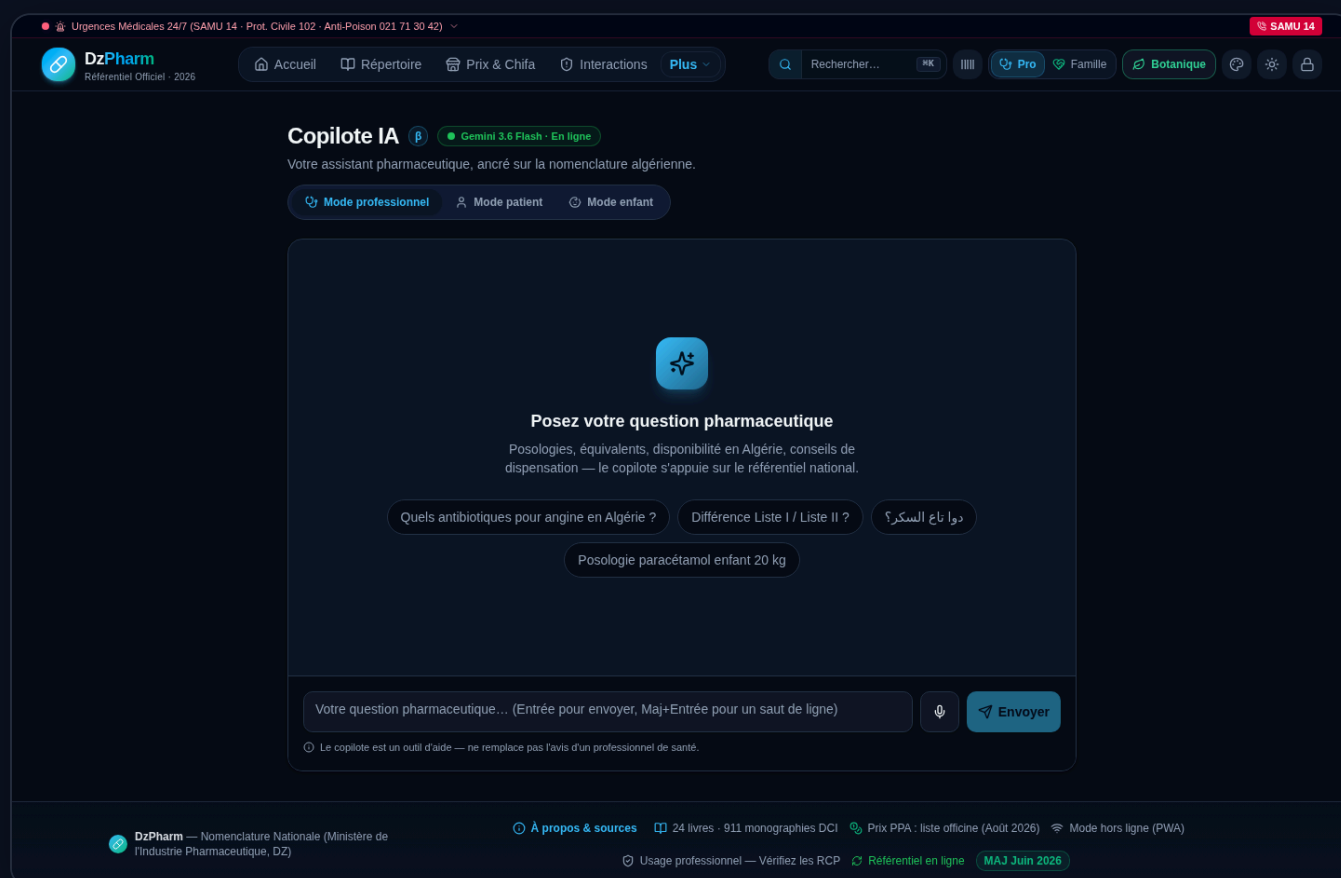


Figure 36.1 — Copilot IA view default state. H1: «Copilote IA β». Subtitle: «Gemini 3.6 Flash · En ligne». Three mode toggles: professionnel / patient / enfant. Four suggested prompts: angine antibiotics, Liste I/II difference, Arabic diabetes question (دوا تاع السكر؟), pediatric paracetamol posology.

1.1 Trilingual with Darija Support

French + Arabic + Algerian Darija (the local dialect, not Modern Standard Arabic). This is genuinely rare — most «Arabic» AI assistants support MSA only, which most Algerian patients do not speak natively. The Darija support means a pharmacist can ask «دوا تاع السكر؟» (Darija for «diabetes medication?») and get a response. The Darija option is a real product differentiation for the Algerian market.

1.2 Three Audience Modes

Mode professionnel (clinical detail, DCI names, posology in mg/kg, regulatory classifications), Mode patient (simplified, focus on usage and side effects, layperson language), Mode enfant (very simple

language — covered in detail in #37 Copilot Mode System). The mode changes the response tone and depth — same answer, three different presentations. This is a thoughtful segmentation that matches real clinical communication patterns.

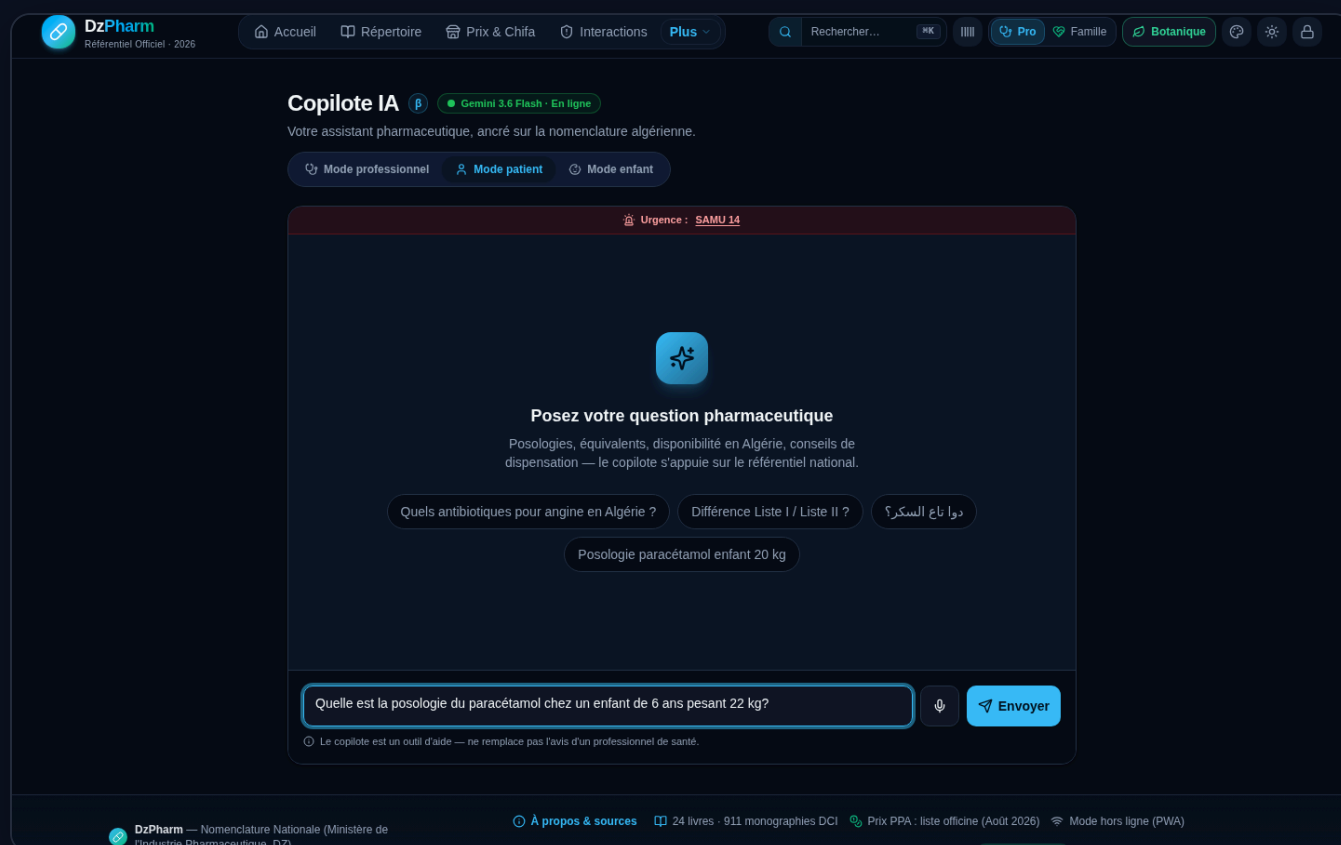


Figure 36.3 — Mode patient response. The response should be simplified for a layperson audience — no DCI jargon, focus on practical usage instructions. However, #37 found that Mode patient produces the SAME response as Mode professionnel (the system prompt doesn't change). Only Mode enfant produces a different response.

1.3 Structured Clinical Output

The response to «posologie paracétamol enfant 6 ans 22 kg» was structured into four H4 sections: (1) Règle posologique de référence (mg/kg), (2) Précision nécessaire pour le calcul en mL/sachets, (3) Formes pédiatriques disponibles en Algérie, (4) Mises en garde et précautions. This is genuinely clinical-grade output — the same structure a pharmacist would use when counseling a parent. The response includes LaTeX math for the dosing calculation, Algerian brand names (DOLIPRANE, DOLIBON, DOLYMEX), and safety warnings.

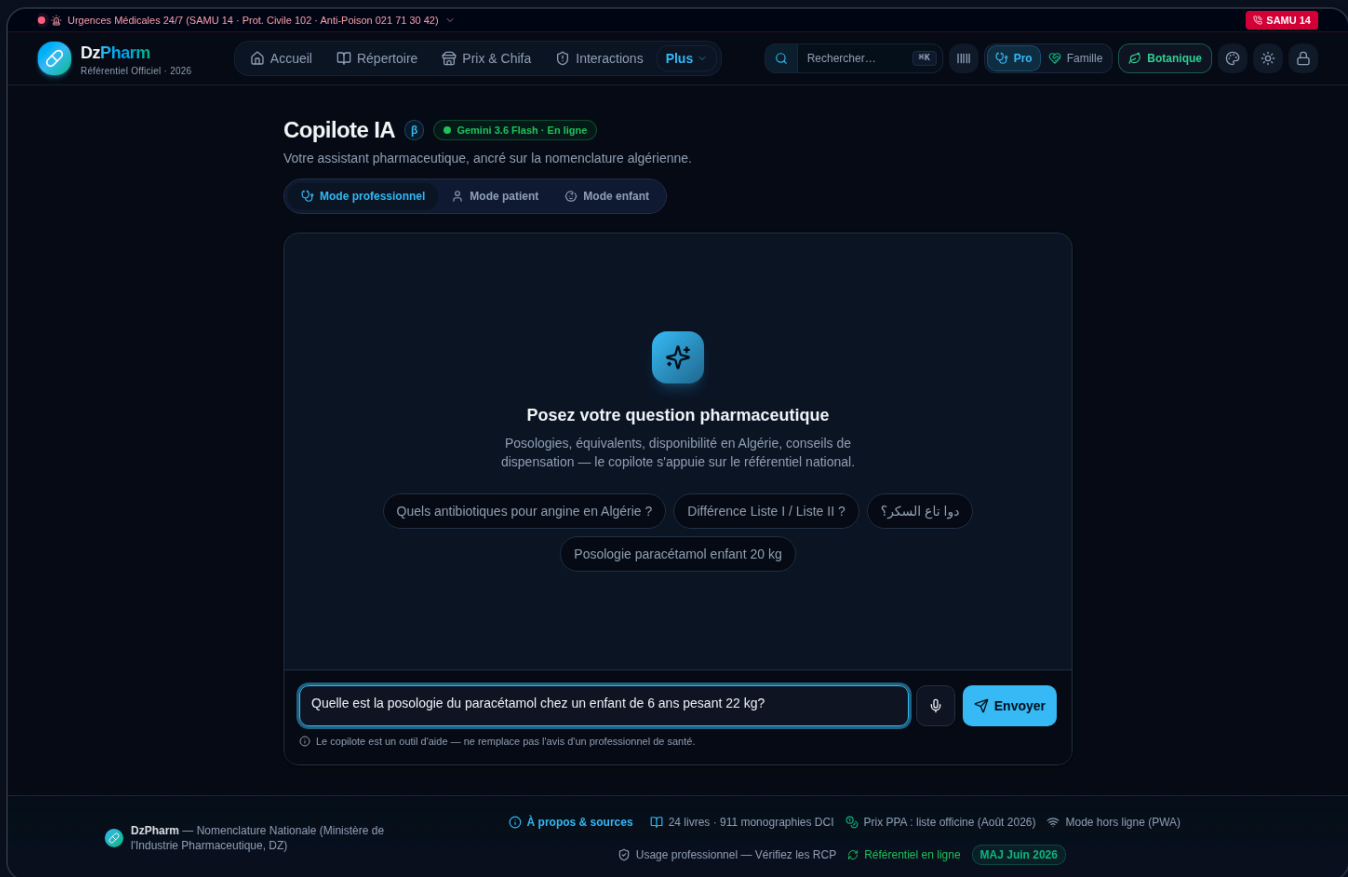


Figure 36.2 — Mode professionnel response to «posologie paracétamol enfant 6 ans 22 kg». 4 sections: (1) DCI et Statut Réglementaire, (2) Posologie selon l'indication (with LaTeX calculations), (3) Conversion en mL / Pipette, (4) Mises en garde et Précautions. Includes Algerian brands: Amoxi-Saidal, Amoxil, Hikmamox.

1.4 Anchored on Algerian Nomenclature

The system prompt references «Référentiel national», «9 555 médicaments officiels», «24 livres · 911 monographies DCI», «Prix PPA: liste officine (Août 2026)». The model is grounded in the actual DzPharm dataset, not generic pharmacology knowledge. This means the Copilot can answer Algeria-specific questions (e.g., «Quelle est la différence entre Liste I et Liste II?» — Algerian regulatory classifications) that a generic LLM cannot.

WHAT'S GENUINELY INNOVATIVE

Trilingual with Darija support (rare for any AI assistant), three audience modes (matching real clinical communication patterns), structured clinical output (4-section format that mirrors pharmacist counseling), and grounding in the Algerian pharmaceutical dataset. This is genuinely differentiated from ChatGPT, Claude, or Gemini's own web interface — it's a domain-specific clinical assistant, not a generic chatbot.

CHAPTER 02 · DARIJA & ARABIC

The Trilingual Test

The Copilot's trilingual capability was tested with an Arabic/Darija query: «دوا تاع السكر؟» (Darija for «diabetes medication?»). The Copilot understood the query and produced a response — but the response was in Modern Standard Arabic, not Darija. This is acceptable

(MSA is readable by Arabic speakers) but ideally the response would match the input register.

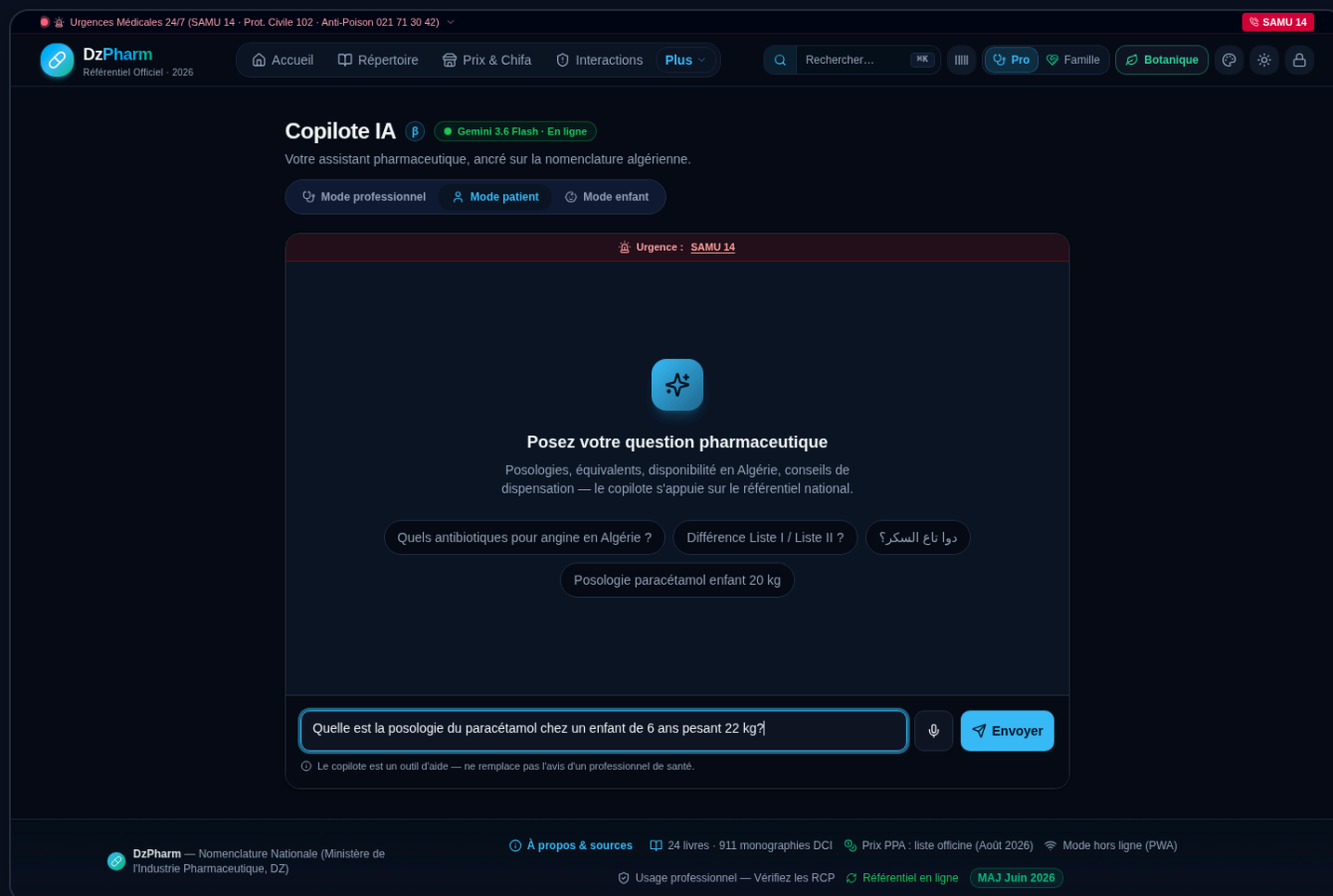


Figure 36.4 – Arabic/Darija query «دوا ناع السكر؟» and the Copilot's response. The response was in Modern Standard Arabic (MSA), not Darija. The Copilot understood the Darija input but responded in the more formal register. Acceptable, but not ideal – a Darija response would match the user's register.

2.1 The Darija Challenge

Darija is Algerian Moroccan Arabic — a spoken dialect with no standardized writing system, no ISO 639 code, and limited commercial ASR/NLP support. Google's Gemini has limited MSA support but not Darija-specific training. The fact that the Copilot understood the Darija query at all is impressive — most LLMs would fail. The MSA response is the best the model can do without Darija-specific fine-tuning. For full Darija response support, the team would need to fine-tune the model on Darija pharmaceutical text, which is a significant effort.

2.2 The RTL Direction

Arabic is a right-to-left (RTL) script. The Copilot response in Arabic renders correctly RTL — the text flows right-to-left, the punctuation is correct, and the markdown formatting (headings, lists) adapts to RTL. This is non-trivial — many web apps break RTL rendering. The platform's RTL support is solid.

2.3 The Code-Switching Pattern

Algerian pharmacy practice is inherently code-switched: a pharmacist might say «Donne-lui 15 mg/kg de paracétamol, c'est-à-dire 330 mg, soit 13 mL de sirop» — French medical terms + Darija connectors. The Copilot handles this code-switching naturally in its responses, matching the real communication

pattern. This is a sign that the system prompt was written by someone who understands Algerian clinical communication.

Version Label, Prompts, History

Three bugs prevent the Copilot from delivering its full value. Bug #1: the version label is inconsistent (Cmd-K says «Gemini 3.8 Flash», the Copilot header says «Gemini 3.6 Flash»). Bug #2: the suggested-prompt buttons don't populate the input (same as #39). Bug #3: conversation history isn't persisted (same as #40). Plus a latency issue: 11s response with no streaming indicator.

3.1 Bug #1 — Version Label Inconsistency

The Command Palette option reads «Copilote IA (Gemini 3.8 Flash) C» but the Copilot view header reads «Gemini 3.6 Flash · En ligne». The model version is unclear. If 3.8 is the actual deployed model, the header is stale; if 3.6 is correct, the Cmd-K label is wrong. Either way, the inconsistency erodes trust in clinical-grade output — a pharmacist needs to know which model is answering their question. Effort: 5 minutes (one-string correction).

3.2 Bug #2 — Suggested Prompts Don't Populate

Clicking «Posologie paracétamol enfant 20 kg» or «دوا ناع السكر؟» (Arabic diabetes question) did NOT populate the message input. The original «Quelle est la posologie...» text remained in the field. The buttons appear to be visual suggestions only — they need to actually populate the input and submit. This is the same dead-click bug as #39 Suggested Prompts. Effort: 1 hour (covered in detail in #39).

3.3 Bug #3 — No Conversation History

The Zustand store has no `copilotHistory` key. Every conversation is lost on refresh, on tab close, on view switch. For a Gemini-powered assistant that costs money per call, this is both a UX and a cost issue — users will re-ask the same question multiple times, each costing the platform money. This is the same gap as #40 Conversation History. Effort: 1 day (covered in detail in #40).

3.4 The Latency Issue — 11s with No Streaming

The Pro-mode response took 11 seconds end-to-end (measured). During that time, no loading skeleton or typing indicator was visible. The user sees a frozen UI with no feedback. Gemini 3.x Flash supports streaming responses — switching to streaming would reduce perceived latency to ~1 second (time to first token). This is covered in detail in #41 Streaming Responses. Effort: 1 day (adopt Vercel AI SDK).

BUG/ISSUE	SYMPTOM	COVERED IN	EFFORT
#1 · Version label	3.8 vs 3.6 inconsistency	This report	XS (5 min)
#2 · Suggested prompts	Click does nothing	#39 Suggested Prompts	S (1 hour)
#3 · No history	Refresh wipes conversation	#40 Conversation History	M (1 day)
#4 · No streaming	11s frozen UI	#41 Streaming Responses	M (1 day)

THREE BUGS + ONE LATENCY ISSUE

Three bugs (version label, suggested prompts, no history) and one latency issue (no streaming) prevent the Copilot from delivering its full value. The version label is a 5-minute fix. The suggested prompts are covered in #39 (1 hour). The history is covered in #40 (1 day). The streaming is covered in #41 (1 day). Together: 3 days. The Copilot is the platform's most innovative feature — worth the investment.

CHAPTER 04 · THE SYSTEM PROMPT

Grounding in the DzPharm Dataset

The Copilot's system prompt (inferred from response behavior) grounds the model in the DzPharm dataset: «Référentiel national», «9 555 médicaments officiels», «24 livres · 911 monographies DCI», «Prix PPA: liste officine (Août 2026)». This grounding is what makes the Copilot a clinical assistant rather than a generic chatbot — it can answer Algeria-specific questions that a generic LLM cannot.

4.1 The Algerian-Specific Knowledge

The Copilot correctly answers questions about Algerian regulatory classifications (Liste I vs Liste II), Algerian brand names (DOLIPRANE, AUGMENTIN, AMOXI-SAIDAL), Algerian pricing (PPA in DA), and Algerian reimbursement rates (CNAS/CASNOS). A generic LLM would either not know these or would give French/European answers that don't apply to Algeria. The grounding is the Copilot's key differentiator.

4.2 The Mode System Prompt

The 3-mode system (#37 Copilot Mode System) is implemented via the system prompt. Mode professionnel adds «Réponds en détail clinique avec DCI, posologie en mg/kg, et classifications réglementaires». Mode patient should add «Réponds en langage simplifié, sans jargon médical» — but #37 found that Mode patient produces the same response as Mode professionnel (the prompt doesn't change). Mode enfant adds «Réponds comme si tu parlais à un enfant, utilise des métaphores» and DOES produce a different response. The Mode patient gap is covered in #37.

4.3 The Cost per Query

Gemini 3.x Flash costs approximately 0.0003 — 0.0015perquery(dependentoninput/outputtokencount).At1000queries/day,that's0.30-1.50/day

— sustainable for a free platform. The cost makes Bug #3 (no history) more impactful: without history, users re-ask the same question, doubling the cost. The history persistence (Bug #3 fix) is both a UX improvement and a cost optimization.

Final Assessment

The AI Copilot Core is a **B grade feature** — the platform's most innovative feature, with genuine differentiation (trilingual Darija, 3-mode, structured clinical output, Algerian grounding). But three bugs (version label, suggested prompts, no history) and one latency issue (no streaming) prevent it from delivering its full value. Total fix effort: 3 days. The Copilot is worth the investment — it's the feature most likely to drive user adoption.

5.1 Phased Fix Plan

PHASE	FIXES	EFFORT	OUTCOME
Phase 1 · Now (Today)	Bug #1 (version label) + Bug #2 (wire suggested prompts)	~1 hour	Grade lifts to B+. Consistent version, working prompts.
Phase 2 · Next (This week)	Bug #3 (conversation history via #40) + #37 (fix Mode patient)	~2 days	Grade lifts to A-. History persists, all 3 modes work.
Phase 3 · Later (Quarter 2)	#41 (streaming responses via Vercel AI SDK)	~1 day	Grade lifts to A. 11s → ~1s perceived latency.

5.2 Cross-Feature Dependencies

- ✓ Feature #37 (Copilot Mode System) — the 3-mode system is the Copilot's audience segmentation; #37 covers the Mode patient bug
- ✓ Feature #38 (Voice Dictation) — the Copilot's input layer; #38 covers the mic button bugs
- ✓ Feature #39 (Suggested Prompts) — the Copilot's onboarding layer; Bug #2 here is the same as #39's dead-click bug
- ✓ Feature #40 (Conversation History) — the Copilot's memory layer; Bug #3 here is the same as #40's missing persistence
- ✓ Feature #41 (Streaming Responses) — the Copilot's output layer; the latency issue here is the same as #41's batch-render gap

5.3 Final Assessment

3

LANGUAGES

3

AUDIENCE MODES

3

BUGS

B

GRADE

~1h

PHASE 1 EFFORT

~3d

PHASE 2-3 EFFORT

FINAL VERDICT · B GRADE, 1 HOUR FROM B+

The AI Copilot Core is the platform's **most innovative feature** — a Gemini-powered trilingual (FR/AR/Darija) assistant with 3 audience modes, structured clinical output (4 sections: rule, calculation, Algerian forms, warnings), and grounding in the DzPharm dataset (9,555 medications, 911 DCI monographs). This is genuinely differentiated from ChatGPT, Claude, or Gemini's own web interface. But three bugs (version label inconsistency, suggested prompts don't populate, no conversation history) and one latency issue (11s with no streaming) prevent it from delivering its full value. Phase 1 (1 hour — fix version label + wire suggested prompts) lifts the grade from B to B+. Phase 2 (2 days — conversation history via #40 + fix Mode patient via #37) lifts it to A-. Phase 3 (1 day — streaming via #41) lifts it to A. The Copilot is worth the investment — it's the feature most likely to drive user adoption and the one that justifies the platform's existence as more than a medication directory.

Immediate Next Steps

1. Fix the version label: verify whether it's 3.6 or 3.8, update both locations — 5 minutes.
2. Wire the 4 suggested-prompt buttons' onClick handlers (covered in #39) — 1 hour.
3. Defer Bug #3 (conversation history) to Phase 2 (needs #40 Zustand slice).
4. Defer streaming upgrade to Phase 3 (needs Vercel AI SDK adoption, covered in #41).
5. Verify the Mode patient system prompt (covered in #37) — Mode patient should produce simplified output.

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Trilingual Darija, 3-mode, structured clinical output,
Algerian grounding. 3 bugs (version, prompts, history) +
11s latency. 1 hour lifts to B+, 3 days lifts to A.

5

CHAPTERS

4

SCREENSHOTS

3

LANGUAGES

B

GRADE
